EXCEL ENGINEERING COLLEGE (AUTONOMOUS) - DEPARTMENT OF ECE

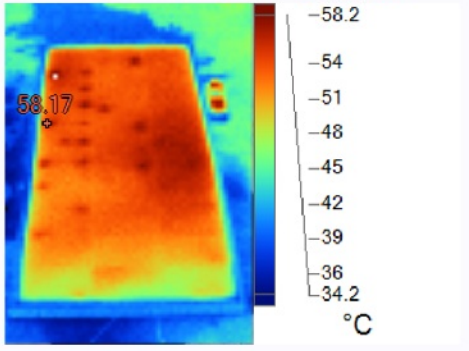
SOLAR PANEL DEFECT DETECTION & DIAGNOSTIC REPORT (MATLAB FUSION PIPELINE)

Panel ID: Panel1 Shadow | Inspection Date: 15-Oct-2026 10:10:35 | Analysis Method: Dual Thermal-Visual Feature Fusion

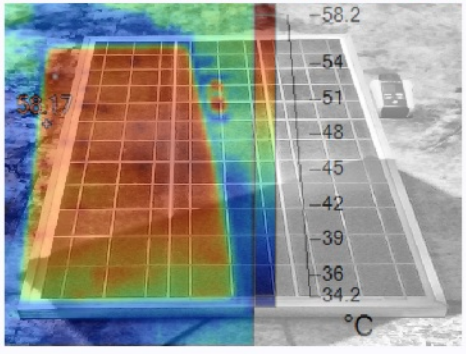
(A) Visual (RGB) Inspection



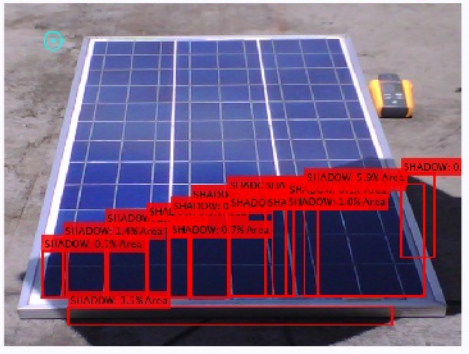
(B) Thermal Thermography (T_{max}: 58.1 °C)



(C) imfuse Thermal-Visual Composite



(D) Defect Localization & Bounding Boxes



DIAGNOSTIC SUMMARY & THERMAL METRICS

- Primary Defect Identified : Shadow / Shading Mismatch
- Peak Temperature (T_{max}) : 58.10 °C (Ambient Mean: 49.91 °C)
- Temperature Rise (Δ T) : +8.20 °C
- Affected Panel Surface : 22.6 %
- Severity Rating : [HIGH]

STATUS HIGH

ENGINEERING MAINTENANCE RECOMMENDATION:

Significant partial shading detected causing reverse bias heating. Remove physical obstructions and verify bypass diodes.